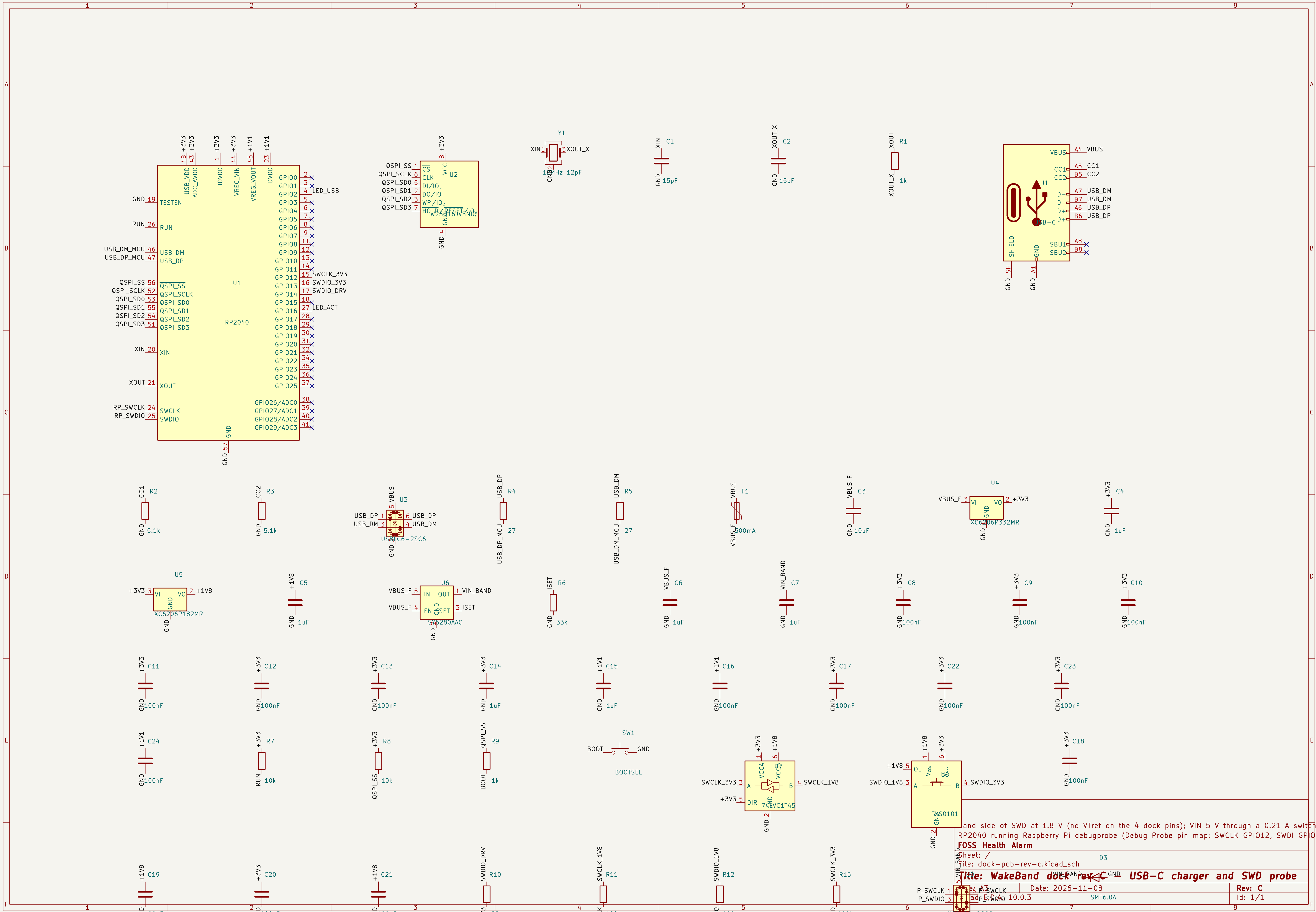

U1
RP2040
+3V3
+1V1
GND
TESTEN
RUN
USB_DM_MCU
USB_DP_MCU
USB_DM
USB_DP
QSPI_SS
QSPI_SCLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
XIN
XOUT
RP_SWCLK
RP_SWDIO
SWCLK
SWDIO
IOVDD
VREG_VIN
VREG_VOUT
DVDD
USB_VDD
ADC_AVDD
LED_USB
SWCLK_3V3
SWDIO_3V3
SWDIO_DRV
LED_ACT
GPIO26/ADC0
GPIO27/ADC1
GPIO28/ADC2
GPIO29/ADC3
U2
W25Q16JVSNIQ
CS
CLK
DI/IO0
DO/IO1
WP/IO2
HOLD/RESET/IO3
VCC
Y1
12MHz 12pF
XOUT_X
C1
15pF
C2
15pF
R1
1k
J1
USB-C
VBUS
CC1
CC2
D−
D+
SBU1
SBU2
SHIELD
R2
5.1k
R3
5.1k
U3
USBLC6-2SC6
R4
27
R5
27
F1
500mA
VBUS_F
C3
10uF
U4
XC6206P332MR
VI
VO
C4
1uF
U5
XC6206P182MR
+1V8
C5
1uF
U6
SY6280AAC
IN
OUT
EN
ISET
VIN_BAND
R6
33k
C6
1uF
C7
1uF
C8
100nF
C9
100nF
C10
100nF
C11
100nF
C12
100nF
C13
100nF
C14
1uF
C15
1uF
C16
100nF
C17
100nF
C22
100nF
C23
100nF
C24
100nF
R7
10k
R8
10k
R9
1k
SW1
BOOTSEL
BOOT
74LVC1T45
VCCA
VCCB
A
B
DIR
SWCLK_1V8
U8
TXS0101
OE
SWDIO_1V8
C18
100nF
C19
100nF
C20
100nF
C21
100nF
R10
22
R11
100
R12
100
R15
100k
P_SWCLK
P_SWDIO
D3
SMF6.0A
USBLC6-2SC6

and side of SWD at 1.8 V (no VTref on the 4 dock pins); VIN 5 V through a 0.21 A switch; ESD/TV
RP2040 running Raspberry Pi debugprobe (Debug Probe pin map: SWCLK GPIO12, SWDI GPIO13, SWD

FOSS Health Alarm

Sheet: /
File: dock-pcb-rev-c.kicad_sch

Title: WakeBand dock rev C – USB-C charger and SWD probe

Size: A3
Date: 2026-11-08
Rev: C
KiCad E.D.A. 10.0.3
Id: 1/1